

Thinking Ahead Without Calling It Proof

A guide to the operational (3,4) Halo(0) experiment

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Central idea: let the program explore coherent hypothetical proof futures as aggressively as useful, but never confuse an imagined future with mathematics that has actually been certified. In short: **imagine freely; verify rigidly.**

This guide explains what the pair **(3,4)** means, what **FUTUREBANK** is, why the design is compared to a chess engine, what the external verifier does, and what has and has not been established by the Halo(0) experiments. It is written to be readable without prior training in automated theorem proving or mathematical logic.

Status note. The experiments described here are research tests. This document does *not* claim that Halo(0) has been formally proved by Predator. A proof claim requires an emitted certificate that survives independent Metamath verification.

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1. What kind of machine is this?

An ordinary automated theorem prover (ATP) tries to construct a formal proof. The broader architecture motivating this project is an **Automated Mathematical Logic Decider (AMLD)**: instead of treating every question as "prove it or fail," an AMLD is designed around the larger idea of **settlement**. Depending on the formal setting, a target statement might eventually be proved, refuted, shown independent of the background assumptions, or exposed as sitting inside an inconsistent assumption set.

The present Halo(0) experiment is narrower: it is exercising the **proof-search side** of that architecture against a precise Metamath target. So it is fair to think of Predator here as an ATP living inside a larger AMLD research program.

What is Halo(0)?

In nonstandard analysis, the hyperreal numbers contain ordinary real numbers together with infinitesimal numbers - nonzero quantities smaller in absolute value than every positive ordinary real threshold of the form $1/n$. **Halo(0)** is the set of infinitesimals around zero. The benchmark statement says, roughly, that this entire halo has the **same cardinality** as the real line. In set-theoretic notation, the target is an equipollence: $\text{Halo}(0) \sim \mathbb{R}$.

Why use it as a benchmark?

It is mathematically meaningful, formally expressible, and difficult enough that blind search cannot plausibly succeed by trivial pattern matching. Most importantly, the experimental policy withholds the stored target proof from the search. The program sees the formal language, the earlier library, and the target statement - not the human route that settles it.

The point of the experiment is not: "Can software replay a proof it was handed?"
The point is: "Can a formally constrained search system discover a verifier-acceptable route while the known target proof is withheld?"

The mathematical library used in the current experiment contains 49,265 pre-target logical assertions. Candidate proofs are expressed as ordinary Metamath proof certificates. Metamath is not part of Predator; it is the independent checker at the end of the line.

2. What does (3,4) mean?

The pair (r,i) is an engineering coordinate. The first coordinate r describes the strength of checked metacontrol or operational reflection. The second coordinate i describes the depth of **reasoned imagination** available to the hypothetical look-ahead mechanism.

The (r, i) coordinate: control vs. imagination

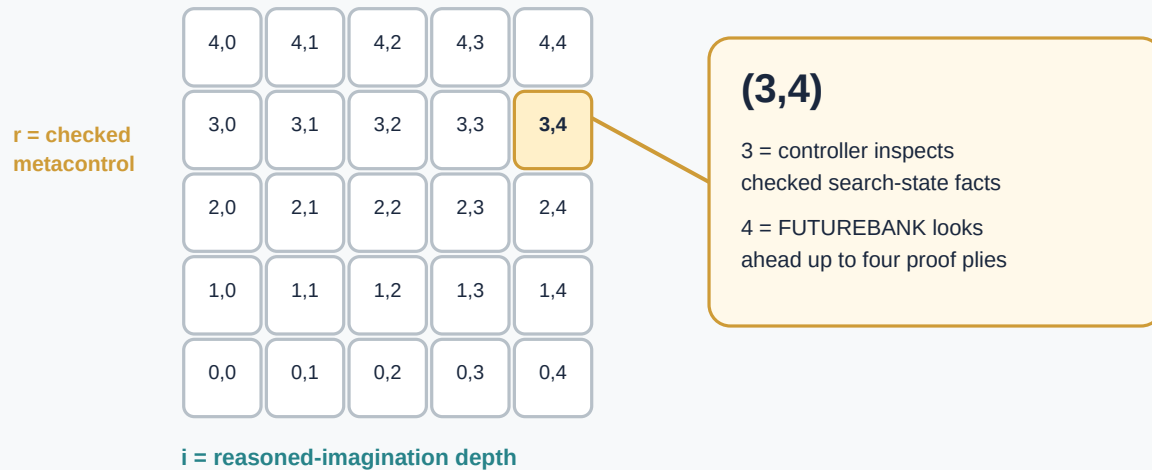


Figure 1. The current coordinate is (3,4): operational level-3 metacontrol paired with up-to-four-ply reasoned imagination.

The 4 is the imagination number. In the current implementation, I4 can simulate up to four further legal backward Metamath assertion applications from a candidate proof state. It is deliberately bounded. It is not "unlimited creativity," and it does not turn imagined steps into facts.

The 3 is the controller number. R3 watches checked facts about the live search - for example whether the best settlement-distance estimate has improved, how long the search has stagnated, whether duplicate states are rising, and whether attractive terminal routes keep failing certificate conditions. It can then change how resources are being used.

Important distinction: (3,4) is not a claim that the machine possesses human consciousness or human self-awareness. It is an operational label for what the software can check about its own search and how deeply it can simulate hypothetical proof continuations.

3. FUTUREBANK: where imagined mathematics is quarantined

The easiest way to understand FUTUREBANK is the chess-engine analogy. A chess engine may internally examine "If I move the rook here, then suppose my opponent replies there, then I respond this way..." None of those imagined moves has actually been played on the board. They are **counterfactual futures** used to choose what to do next.

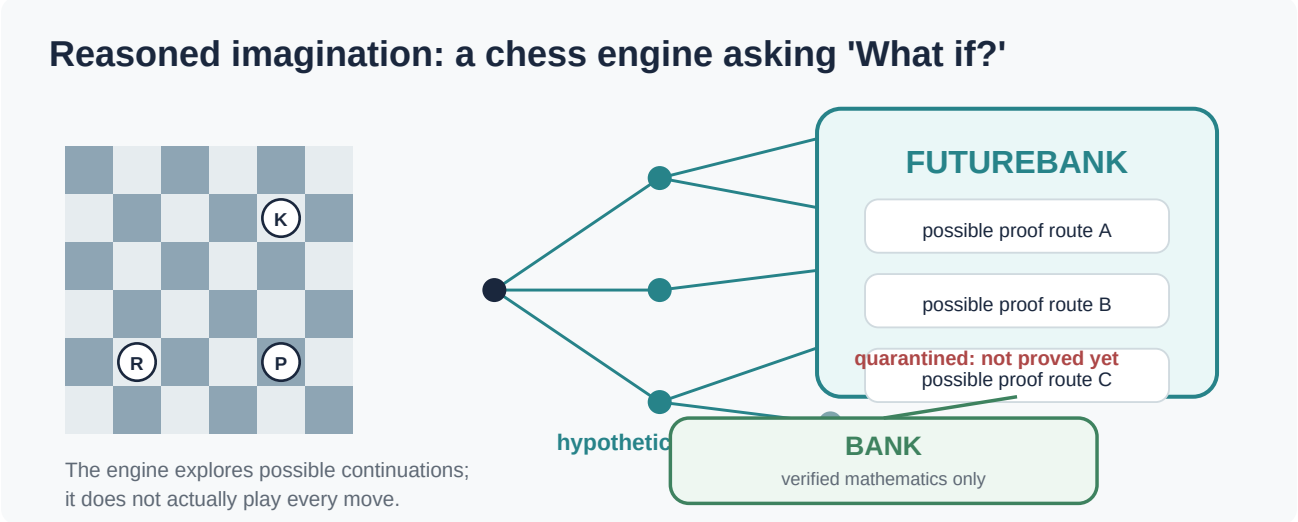


Figure 2. FUTUREBANK is the hypothetical workspace. BANK is the verified record.

That is essentially the role of FUTUREBANK. Statements and proof states generated by the **I4 reasoned-imagination mechanism** live there conceptually as hypotheticals. The system can score them, compare them, notice that some routes become promising, and use them to rank the next real search step. But FUTUREBANK is a **quarantine**: an imagined statement is not a theorem merely because it is coherent or useful.

The **R3 controller** does something different. It decides how strongly to rely on those hypothetical forecasts and when the current search strategy has become unproductive. In other words: **I4 generates the look-ahead; R3 governs the search that uses it.**

BANK vs. FUTUREBANK

BANK	Verified shared mathematical memory. A statement belongs here only after the formal verification boundary has been crossed.
FUTUREBANK	Hypothetical proof futures. They may be coherent, rooted in legal proof operations, and useful for ranking - but they remain unproved until certified.
Why separate them?	Because the separation prevents a plausible idea from being silently promoted into mathematics. This is the architectural version of "imagine freely; verify rigidly."

4. Why imagination is useful - and dangerous

A theorem prover that never looks ahead can be shortsighted. It may repeatedly choose a locally attractive step that opens a terrible set of subgoals. Reasoned imagination gives the search a limited way to ask: *If I take this step, what does the next small piece of the proof landscape look like?*

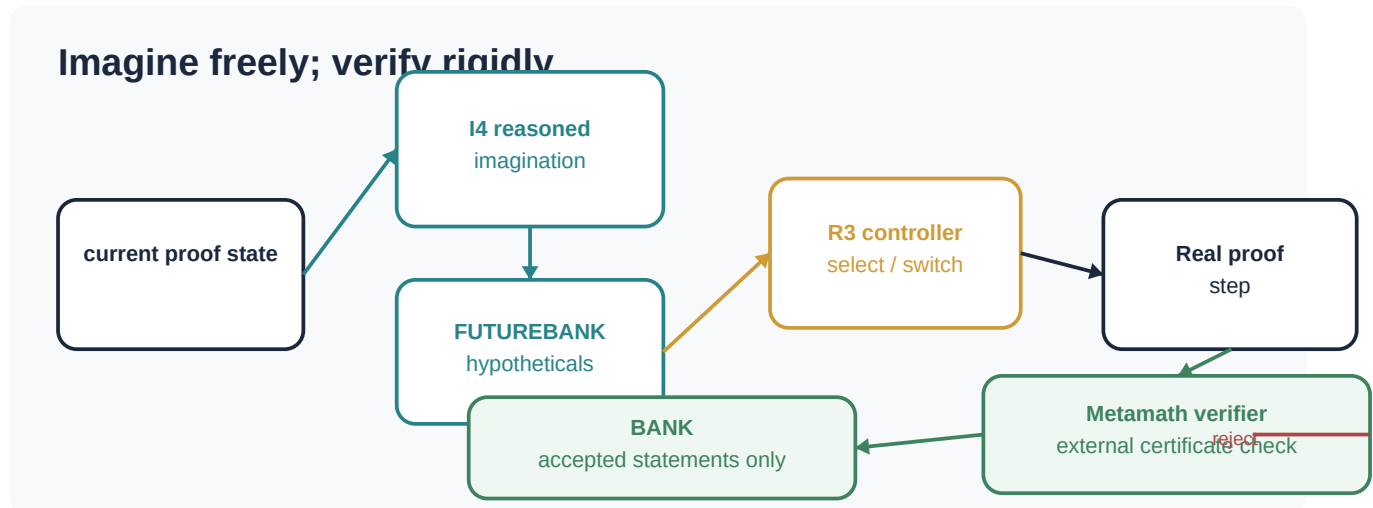


Figure 3. Only real proof steps can eventually reach the external verifier and then BANK.

But there is a trap. A hypothetical route can be syntactically elegant and still be impossible to certify. Metamath, for example, enforces **disjoint-variable (DV) conditions**. Earlier experiments repeatedly found complete-looking short routes that collapsed only when unresolved variables were finally grounded: variables that were required to stay distinct became the same variable.

That failure was useful. The engineering response was not to weaken the verifier. It was to make the search and the imagination more certificate-aware. The real search now prunes forced DV violations, checks terminal branches after the exact grounding used for certificate emission, and continues searching when a complete-looking route is illegal. The I4 FUTUREBANK now carries the accumulated DV obligations too, so an imagined zero-goal future receives a "solved" bonus only if it would survive the same terminal certificate condition.

This is the epistemic boundary: plausibility can guide search; legality can prune search; but only a verifier-accepted certificate establishes the theorem.

5. What the 3 actually does: change strategy when search saturates

A common failure mode in proof search is **saturation**: the system keeps doing more of the same work without getting meaningfully closer to a certificate. In the Halo experiments this became visible as long stretches with no improvement in the best settlement-distance estimate while large numbers of FUTUREBANK states were still being simulated.

Operational R3 responds with distinct search modes:

COMPASS	Normal settlement-distance-guided search with selective I4 look-ahead.
CERTIFY	After stagnation or repeated illegal terminal routes, penalize unresolved certificate risk and reduce imagination breadth.
DIVERSIFY	Sample a broader set of proof directions and use imagination more sparsely, trying to escape the current basin.
LEAN	When saturation is severe, spend most of the wall clock on real proof expansions and only a thin I4 probe.

A genuine improvement resets the saturation clock. The controller can return toward COMPASS rather than remaining permanently in an emergency mode. When strategy changes, surviving frontier states are re-ranked rather than simply thrown away.

Predator 8.008: the population finally becomes wall-clock fair

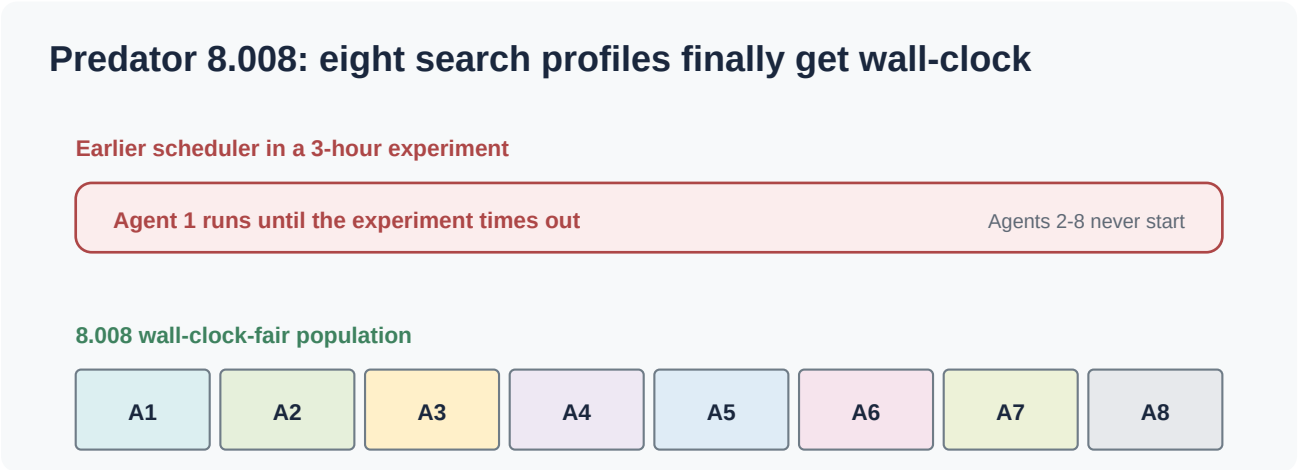


Figure 4. The current scheduler guarantees bounded wall-clock access to all eight profiles.

The earlier population scheduler divided the *expansion* budget fairly but ran agents sequentially. On Halo, the first agent could not exhaust its multi-million-expansion share within three hours, so agents 2 through 8 never started. Version 8.008 changes only population scheduling: each of eight search profiles receives a bounded wall-clock slice. The current internal population clock is 10,720 seconds (178 minutes 40 seconds), giving an initial slice of 1,340 seconds (22 minutes 20 seconds) per profile. If an agent finishes early, unused time is redistributed forward. Expansion shares remain hard caps whose sum is still 50,000,000.

6. What the experiments have actually taught us

The strongest value of these runs so far is diagnostic. Several apparent "mathematical failures" turned out to be engineering failures, and each one sharpened the architecture.

Indexing	The first (3,4) run never reached proof search because the generic parser spent the entire window preprocessing the large library. The indexer was rebuilt so the full pre-target corpus could be cached without silently omitting pathological formulas.
Illegal near-proofs	Early searches found complete-looking routes quickly, but Metamath rejected them for DV violations. Terminal-ground DV checking was added so the search rejects those shortcuts and continues instead of stopping.
Imagination saturation	A 50-minute run showed hundreds of thousands of imagined states with little object-level progress. R3 strategy switching was added to reduce imagination when it becomes expensive and repetitive.
Certificate-coherent imagination	FUTUREBANK was upgraded so imagined states carry the same accumulated DV constraints as real search. Imagined "solutions" that would fail certificate grounding are now rejected inside imagination.
Population scheduling	A three-hour run revealed that only the first of eight agents had actually received wall-clock time. 8.008 now slices time fairly across all eight profiles.

What has *not* been shown

No completed experiment described here justifies saying that operational (3,4) has proved the Halo(0) theorem. No finite timeout shows the theorem is false or unprovable. No synthetic success shows that the full Halo benchmark must succeed. And an attractive search-state score is not itself a mathematical result.

What *has* been shown is narrower and useful: the machinery can generate nontrivial candidate routes, detect several classes of certificate failure, alter strategy in response to stagnation, make its imagined futures more certificate-coherent, and now allocate the finite wall clock across genuinely different search profiles.

Research question now: Can the combination of checked metacontrol (R3), bounded reasoned imagination (I4), certificate-aware hypothetical search, and a wall-clock-fair population reach a verifier-clean Halo certificate before the finite resource bounds are exhausted?

7. The whole idea in one page

Start with a formal target. Predator is given a precise statement and a large formal library, but the benchmark policy withholds the target's stored proof from search.

Search real proof states. Every real step must be a legal backward use of an earlier Metamath assertion.

Look ahead hypothetically. I4 can simulate up to four further legal proof applications. Those counterfactual states belong conceptually to FUTUREBANK, not BANK.

Use checked metacontrol. R3 watches objective search signals and can change resource strategy: COMPASS, CERTIFY, DIVERSIFY, or LEAN.

Reject impossible futures early. Real and imagined branches carry certificate constraints, including Metamath disjoint-variable requirements.

Never promote imagination by assertion. A route matters mathematically only if a concrete certificate is emitted and accepted by the external verifier.

Compact glossary

ATP	Automated theorem prover: a program that searches for formal proof certificates.
AMLD	Automated Mathematical Logic Decider: the broader settlement architecture in which proving is one possible terminal status.
(3,4)	Operational coordinate: R3 checked metacontrol plus I4 four-ply reasoned imagination.
BANK	Verified shared mathematical memory.
FUTUREBANK	Quarantined hypothetical proof futures used for look-ahead and ranking.
Settlement distance	A practical heuristic estimate of how far a search state appears to be from a settlement; useful for control, not itself a proof.
Metamath	The independent formal proof checker used to accept or reject emitted certificates.

Public links and technical sources

Brian Tenneson email: btenneson2301@baypath.edu
Substack: <https://btenneson2301.substack.com>
ATP / Predator repository: <https://github.com/btenneson/ATP>

Implementation discussed here: Predator 8.008-R3I4-wallfair-population.py, building on 8.007 DV-coherent imagination, 8.006 strategy switching, and the earlier full-index / DV safeguards. The current workflow fixes the Halo target, the frozen set.mm baseline, the three-hour shell clock, the eight-profile population, and the independent verification boundary.

Final standard: If Predator ever emits a Halo candidate, the relevant question is not whether the path looked intelligent, imaginative, or close. The relevant question is whether Metamath accepts the certificate.

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